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# A parametric study of layered bed PSA for hydrogen purification

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ABSTRACT

The production of high purity hydrogen (99.99+%) at reduced cost is an important and sought target. This work is focused on the separation of hydrogen from a five component mixture ($H_2/CO_2/CH_4/CO/N_2$) by pressure swing adsorption. A complete mathematical model that describes the dynamic behaviour of a PSA unit is presented. This model is applied in the study of the behaviour of both single column and four columns PSA processes with layered activated carbon/zeolite beds and with an eight steps cycle. In the single column simulation, a 99.9994% purity hydrogen stream is attained at the end of the feed step for a process hydrogen recovery of 51.84% and a productivity of 59.6 $mol_{H_2}/kg_{ads}$/day. The multicolumn simulation predicts a hydrogen recovery and purity, respectively, of 52.11% and 99.9958%. The influence of feed flow rate, purge to feed ratio and lengths of both adsorbent layers on the system performance is assessed. It is shown that the introduction of the zeolite layer improves both the purity and recovery of the process. Reduced models are formulated based on the sequential identification of controlling resistances in the complete model. The predictions of the reduced models are evaluated by comparing their results with those obtained from the complete model. It is shown that the model that merely takes into account the micropore resistance (described by the LDF model) and assumes thermal equilibrium only between the gas and solid phases satisfactorily predicts the behaviour of the pressure swing adsorption unit.

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## 1. Introduction

Hydrogen is used in many important applications in the chemical process industries, such as, hydrocracking, hydrogenation of fats and oils, methanol production, manufacture of silicon, etc. Due to environmental concerns, the use of hydrogen for fuel cell has also received much attention lately.

The most economical way to produce hydrogen is the catalytic reforming of natural gas combined with a water gas shift reaction step, from which a hydrogen stream containing several impurities, namely water vapour, carbon dioxide, methane, carbon monoxide and nitrogen, is obtained. These impurities must be removed in order to attain the required high purity $H_2$ (99.99+%) for fuel cell applications. This separation is usually performed with a pressure swing adsorption (PSA) unit (Ruthven et al., 1994; Yang, 1987). An increase in the demand for hydrogen has promoted a strong economic motivation for the development of new improved PSA processes, namely the development of multibed processes (Batta and Island, 1971; Fuderer and Rudelstorfer, 1976; Yamaguchi and Kobayashi, 1993), the application of pressure equalization steps, the combination of different adsorbents (Bomard et al., 1997; Golden et al., 1990; Le Bec, 2005; Plee, 2002) and more recently the development of rapid PSA processes (Golden et al., 2006).

The use of multiple adsorbents, placed as layers in the beds, is an important characteristic of the currently employed PSA technology for hydrogen purification. As the number of gases to be removed from the stream is considerable, the different layers of adsorbents are used to remove a particular contaminant or group of contaminants. Commonly the first layer is either alumina or silica that will preferably retain the water vapour. Then an activated carbon layer is used to selectively adsorb carbon dioxide and hydrocarbons such as methane. The last layer is a zeolite with enhanced capacity for carbon monoxide and nitrogen. $CO_2$ and water vapour are strongly adsorbed in the zeolite and cannot be readily desorbed by decreasing the pressure and start to accumulate in this adsorbent as the cycles proceed. This imposes an additional constraint in the design of the layered PSA, that is, besides the fact that the lighter impurities ($CH_4$, CO and $N_2$) cannot break through the entire bed, $CO_2$ and water vapour must be prevented from reaching the zeolite layer (Chlendi and Tondeur, 1995).

There are different publications concerning the study of the dynamic behaviour of layered beds (Jee et al., 2001; Park et al., 1998) and layered PSA (Ahn et al., 1999, 2001; Park et al., 2000; Yang et al., 1995; Yang and Lee, 1998; Zhou et al., 2002) for the purification of

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hydrogen with the aim of getting a better understanding of the system and also seeking for optimal (or at least increased) performance. Nevertheless, the production cost of hydrogen should be reduced to implement it as an energy vector, either in energy production or as fuel, and thus, further improvements that reduce the capital and operating costs are yet crucial. These performance improvements can be achieved by the development of superior adsorbents, by changing the PSA operating scheme, as well as, by the optimization of the operation parameters of the process.

In this study, hydrogen purification by a layered activated carbon/zeolite PSA process is considered. A cycle scheme with eight steps, including two pressure equalizations is presented. A complete mathematical model that takes into account multicomponent adsorption, velocity variations inside the column, non-isothermal behaviour and both external and internal (micropore and macropore) resistances to diffusion is presented. Single column simulations were performed to assess the general behaviour of the system and also the effect of feed flow rate, purge to feed ratio, length of activated carbon layer at constant total length, length of activated carbon at constant zeolite length and length of zeolite at constant activated carbon length on the system performance. In order to improve the performance of the process an optimization study is needed which is an extremely time consuming task. The use of simplified models which need less computational time are desirable to facilitate the optimization study. The simplified model should be capable to satisfactorily predict the PSA behaviour. Based on the sequential identification of controlling resistances in the complete model, two different simplifications are discussed and two reduced models are obtained. The reduced model predictions are evaluated by comparing their results with the ones obtained from the complete model.

Pure hydrogen is used in the purge and pressurization steps of the one-column PSA simulations. In real multibed PSA systems recycled streams are used (in pressurization, purge, etc). In order to evaluate the effect of stream recycling, we compared the results of one-column simulation with the simulation of a four column system.

The objectives of this work are: (i) the implementation of a model reduction strategy that results in shorter computational time; (ii) the evaluation of the reduced model predictions by comparison with the complete model; (iii) to get an insight on the PSA process for purification of hydrogen through a parametric study that assesses the influence of some operating parameters on the system performance.

## 2. Mathematical model

### 2.1. Complete model

In a cyclic adsorption process there are two actions involved: the adsorption, where the more adsorbed species (heavy components) are retained in the adsorbent and the lighter component(s) elute at the column end; and the desorption, where the heavy components are removed from the bed partially regenerating the adsorbent for the next cycle. A PSA cycle is a sequential combination of elementary pre-defined steps where the regeneration is made by reducing the total pressure of the bed. A generic mathematical model employed in the description of the behaviour of a PSA process should couple mass, momentum and energy balances over a packed bed with the appropriate boundary conditions for each of the steps comprising the cycle.

In a fixed bed, three phases can be distinguished: the gas phase, the solid phase where adsorption and diffusion take place and the column wall where energy may be transferred to (or from) the surroundings. A schematic representation of this system is shown in Fig. 1.

The development of a complete mathematical model capable to describe the dynamic behaviour of multicomponent adsorption in a fixed bed was based on the following assumptions:

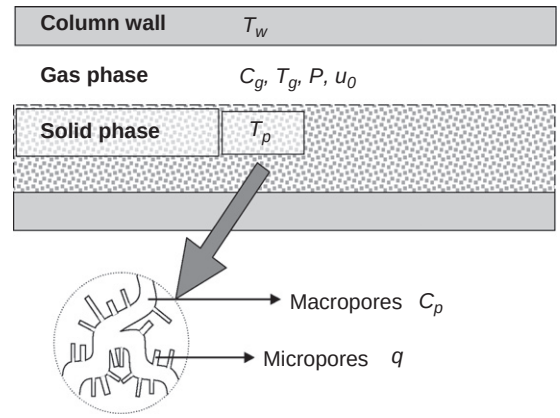


**Fig. 1.** Schematic representation of the adsorption bed.

1. Ideal gas behaviour throughout the column: within the operating conditions under study (up to 7 bar), the gases follow nearly the ideal gas law.
2. No mass, heat or velocity gradients in the radial direction.
3. Axial dispersed plug flow.
4. External mass and heat transfer resistances expressed with the film model.
5. Bidispersed adsorbent particle with macropore and micropore mass transfer resistances, both expressed with the linear driving force (LDF) model.
6. No temperature gradients inside each particle as the heat transfer in the solid particles is much higher than in the gas phase.
7. The column wall interchanges energy with the gas phase inside the column and with the external environment: constant heat transfer coefficients and constant external environment were considered.
8. Constant porosity along the bed.
9. The Ergun equation is valid locally: in the momentum balance, only the terms of pressure drop and velocity change are considered.

The resulting mass, momentum and energy balance equations are presented in Table 1.

In order to solve this system of partial differential equations, boundary and initial conditions are needed. As mentioned before, a PSA cycle is a sequence of elementary steps which require appropriate boundary conditions. The most common steps encountered in PSA processes are pressurization, feed, blowdown, purge, rinse and pressure equalization. The boundary conditions associated with each of these steps are given in Table 2. The pressure equalization steps are frequently used in multicolumn processes to reduce the energy consumption of the system (Ruthven et al., 1994). In the pressure equalization—depressurization step and in the blowdown step, $P_{\text{exit}}$ is considered to decrease exponentially with time during the steps, representing a valve equation. In the case of the equalization—pressurization step, the inlet pressure increases with time such that the molar flow rate ($u_0 C_g$) entering the column is equal to the molar flow rate that exited the column during the depressurization. At the layer interface no additional boundary conditions are used as the model takes care of the change by using, in the balance equations, the appropriate adsorbent properties.

This model has already been used in the simulation of fixed bed behaviour and PSA applications of different mixtures showing very good agreement between predictions and experimental data

**Table 1**
Mass, momentum and energy balance equations of the complete mathematical model of a fixed bed adsorption system

Mass Balances
*Gas phase*:

$$\frac{\partial}{\partial z}\left(\varepsilon D_{ax} C_{g,T}\frac{\partial y_i}{\partial z}\right) - \frac{\partial}{\partial z}(u_0 C_{g,i}) - \varepsilon\frac{\partial C_{g,i}}{\partial t} - \frac{(1-\varepsilon)a_p k_f}{1+Bi_i}(C_{g,i} - \overline{C_{p,i}}) = 0$$

*Solid phase—macropore*:

$$\frac{\partial \overline{C_{p,i}}}{\partial t} = \frac{\Omega_M D_{p,i} Bi_i}{R_p^2(1+Bi_i)}(C_{g,i} - \overline{C_{p,i}}) - \frac{\rho_p}{\varepsilon_p}\frac{\partial \overline{q_i}}{\partial t}$$

*Solid phase—micropore*:

$$\frac{\partial \overline{q_i}}{\partial t} = \frac{\Omega_c D_{c,i}}{r_c^2}(q_i^* - \overline{q_i})$$

Momentum balance

$$-\frac{\partial P}{\partial z} = \frac{150\mu(1-\varepsilon)^2}{\varepsilon^3 d_p^2}u_0 + \frac{1.75(1-\varepsilon)\rho}{\varepsilon^3 d_p}|u_0|u_0$$

Energy balances
*Gas phase*:

$$\frac{\partial}{\partial z}\left(\lambda\frac{\partial T_g}{\partial z}\right) - u_0 C_{g,T} C_p\frac{\partial T_g}{\partial z} + \varepsilon R_g T_g\frac{\partial C_{g,T}}{\partial t} - (1-\varepsilon)a_p h_f(T_g - T_p) - \frac{4h_w}{d_{wi}}(T_g - T_w) - \varepsilon C_{g,T} C_V\frac{\partial T_g}{\partial t} = 0$$

*Solid phase*:

$$(1-\varepsilon)\left[\varepsilon_p\sum_{i=1}^{n}\overline{C_{p,i}}C_{v,i} + \rho_p\sum_{i=1}^{n}\overline{q_i}C_{v,\mathrm{ads},i} + \rho_p\hat{C}_{ps}\right]\frac{\partial T_p}{\partial t} = (1-\varepsilon)\varepsilon_p R_g T_p\frac{\partial \overline{C_{p,T}}}{\partial t} + \rho_b\sum_{i=1}^{n}(-\Delta H_{\mathrm{ads}})_i\frac{\partial \overline{q_i}}{\partial t} + (1-\varepsilon)a_p h_f(T_g - T_p)$$

*Column wall*:

$$\rho_w\hat{C}_{p,w}\frac{\partial T_w}{\partial t} = \alpha_w h_w(T_g - T_w) - \alpha_{w\ell}U(T_w - T_\infty)$$

(Cavenati et al., 2006; Da Silva et al., 1999; Grande and Rodrigues, 2005).

### 2.2. Reduced model

The simulation of a PSA process, until cyclic steady state (CSS) is achieved, takes a considerable amount of CPU time. Due to the large number of variables and parameters, a substantial number of simulations are needed to find the optimal process/operating conditions. To facilitate an optimization study, a reduced model with satisfactory predicting capabilities is needed to decrease the required computational time.

The reduced models were obtained based on the sequential identification of controlling resistances in the complete model presented above, either in the mass or energy transfer. For this reason, two possible simplifications are provided: one within the mass balances and another within the energy balances. The model reduction strategy presented below is based on data obtained for the hydrogen process. However, this same strategy, that evaluates the dominant/controlling phenomena based on time constants analysis, can also be used for other systems.

The simplification in the mass balances considers a limiting case where one of the mass transfer mechanisms controls. This simplification is supported by a previous study (Lopes et al., 2007) where the kinetic properties of pure gases on two commercially available adsorbents were determined. It was established that for the gases under study, micropore diffusion is the controlling mass transfer resistance. For this reason, the reduced model formulation considered that the film mass transfer and macropore diffusion were negligible and only the micropore diffusion is accounted for. With this assumption, the macropore concentration in the pellets is equal to the gas phase concentration and therefore the macropore mass balance is eliminated from the model.

The complete model presented in Section 2.1 includes three energy balances, one for each phase (gas, solid and column wall). If thermal equilibrium is assumed to exist between two or more phases, the model can be further simplified. Preliminary simulation results showed that, at CSS, the temperature profiles of the gas and solid phases were almost identical, while a considerable difference to the column wall temperatures was observed. Based on this observation, two possible simplifications are proposed. The first simplification considers that only the gas and solid phases are in equilibrium and that a thermal gradient to the column wall exists. With these hypotheses, two energy balances must be included in the model (instead of the original three equations of the complete model). With this simplification, one variable is excluded from the model: the temperature of the solid.

The second simplification assumes that all the system is in thermal equilibrium. This means that the column wall temperature is the same as the temperature in the gas and solid phases. In this way, the model is reduced to only one energy balance.

Taking into consideration the simplifications discussed above, the balance equations presented in Table 3 are obtained and two reduced models are formulated: the “two energy balances model” and the “one energy balance model”. With the simplifications imposed,

**Table 2**
Boundary conditions associated with the steps commonly used in PSA operation

| | |
|---|---|
| Co-current pressurization with feed<br>Inlet, $z=0$<br>$u_{0\text{inlet}}C_{\text{inlet},i} = u_0 C_{g,i} - \varepsilon D_{\text{ax}} C_{g,T} \frac{\partial y_i}{\partial z}$<br>$P = P_{\text{inlet}}$<br>$u_{0\text{inlet}}C_{\text{inlet},T} C_p T_{\text{inlet}} = u_0 C_{g,T} C_p T_g - \lambda \frac{\partial T_g}{\partial z}$ | $z=L$<br>$\frac{\partial C_{g,i}}{\partial z} = 0$<br>$u_0 = 0$<br>$\frac{\partial T_g}{\partial z} = 0$ |
| Counter-current pressurization with the less adsorbed compound<br>$z=0$<br>$\frac{\partial C_{g,i}}{\partial z} = 0$<br>$u_0 = 0$<br>$\frac{\partial T_g}{\partial z} = 0$ | Inlet, $z=L$<br>$u_{0\text{inlet}}C_{\text{inlet},i} = u_0 C_{g,T} C_p T_g - \lambda \frac{\partial T_g}{\partial z}$<br>$P = P_{\text{exit}}$<br>$u_{0\text{inlet}}C_{\text{inlet},T} C_p T_{\text{inlet}} = u_0 C_{g,i} - \varepsilon D_{\text{ax}} C_{g,T} \frac{\partial y_i}{\partial z}$ |
| Feed<br>Inlet, $z=0$<br>$u_{0\text{inlet}}C_{\text{inlet},i} = u_0 C_{g,i} - \varepsilon D_{\text{ax}} C_{g,T} \frac{\partial y_i}{\partial z}$<br>$u_{0\text{inlet}}C_{\text{inlet},T} = u_0 C_{g,T}$<br>$u_{0\text{inlet}}C_{\text{inlet},T} C_p T_{\text{inlet}} = u_0 C_{g,T} C_p T_g - \lambda \frac{\partial T_g}{\partial z}$ | Outlet, $z=L$<br>$P = P_{\text{exit}}$<br>$\frac{\partial T_g}{\partial z} = 0$<br>$\frac{\partial C_{g,i}}{\partial z} = 0$ |
| Counter-current blowdown<br>Outlet, $z=0$<br>$\frac{\partial C_{g,i}}{\partial z} = 0$<br>$P = P_{\text{exit}}$<br>$\frac{\partial T_g}{\partial z} = 0$ | $z=L$<br>$\frac{\partial C_{g,i}}{\partial z} = 0$<br>$u_0 = 0$<br>$\frac{\partial T_g}{\partial z} = 0$ |
| Counter-current purge with the less adsorbed compound<br>Outlet, $z=0$<br>$\frac{\partial C_{g,i}}{\partial z} = 0$<br>$P = P_{\text{exit}}$<br>$\frac{\partial T_g}{\partial z} = 0$ | Inlet, $z=L$<br>$u_{0\text{inlet}}C_{\text{inlet},i} = u_0 C_{g,i} - \varepsilon D_{\text{ax}} C_{g,T} \frac{\partial y_i}{\partial z}$<br>$u_{0\text{inlet}}C_{\text{inlet},T} = u_0 C_{g,T}$<br>$u_{0\text{inlet}}C_{\text{inlet},T} C_p T_{\text{inlet}} = u_0 C_{g,T} C_p T_g - \lambda \frac{\partial T_g}{\partial z}$ |
| Co-current rinse with the most adsorbed compound<br>Inlet, $z=0$<br>$u_{0\text{inlet}}C_{\text{inlet},i} = u_0 C_{g,i} - \varepsilon D_{\text{ax}} C_{g,T} \frac{\partial y_i}{\partial z}$<br>$u_{0\text{inlet}}C_{\text{inlet},T} = u_0 C_{g,T}$<br>$u_{0\text{inlet}}C_{\text{inlet},T} C_p T_{\text{inlet}} = u_0 C_{g,T} C_p T_g - \lambda \frac{\partial T_g}{\partial z}$ | Outlet, $z=L$<br>$\frac{\partial C_{g,i}}{\partial z} = 0$<br>$P = P_{\text{exit}}$<br>$\frac{\partial T_g}{\partial z} = 0$ |
| Pressure equalization—depressurization<br>$z=0$<br>$\frac{\partial C_{g,i}}{\partial z} = 0$<br>$u_0 = 0$<br>$\frac{\partial T_g}{\partial z} = 0$ | Outlet, $z=L$<br>$\frac{\partial C_{g,i}}{\partial z} = 0$<br>$P = P_{\text{exit}}$<br>$\frac{\partial T_g}{\partial z} = 0$ |
| Pressure equalization—pressurization<br>$z=0$<br>$\frac{\partial C_{g,i}}{\partial z} = 0$<br>$u_0 = 0$<br>$\frac{\partial T_g}{\partial z} = 0$ | Inlet, $z=L$<br>$u_{0\text{inlet}}C_{\text{inlet},i} = u_0 C_{g,i} - \varepsilon D_{\text{ax}} C_{g,T} \frac{\partial y_i}{\partial z}$<br>$u_{0\text{inlet}}C_{\text{inlet},T} = u_0 C_{g,T}$<br>$u_{0\text{inlet}}C_{\text{inlet},T} C_p T_{\text{inlet}} = u_0 C_{g,T} C_p T_g - \lambda \frac{\partial T_g}{\partial z}$ |

the seven balance equations of the complete model are reduced, respectively, to five equations in the two energy balances model or to four equations in the one energy balance model.

In the next section, simulation results of the complete and of the reduced models are compared in order to assess the validity of the proposed simplifications.

## 3. PSA simulations

The models presented in Section 2 were applied for the simulation of a single column PSA process with eight elementary steps. The cycle sequence considered, which includes two pressure equalizations, is schematically represented in Fig. 2. The extension of the proposed cycle to a four columns process with continuous feed is given in Fig. 3. The duration of the feed step automatically presets the duration of the cycle and also of the other steps through the relations showed in Fig. 3. These relations were taken into consideration even in the single column simulations.

The mathematical models were implemented in gPROMS environment (Process System Enterprise, London, UK) and numerically solved using the centred finite difference method, with 180 discretization intervals and second order approximation for partial

**Table 3**
Mass, momentum and energy balance equations of the reduced mathematical model of a fixed bed adsorption system

Mass balances

$$\frac{\partial}{\partial z}\left(\varepsilon D_{ax} C_{g,T}\frac{\partial y_i}{\partial z}\right) - \frac{\partial}{\partial z}(u_0 C_{g,i}) - \varepsilon\frac{\partial C_{g,i}}{\partial t} - (1-\varepsilon)\left(\varepsilon_p\frac{\partial C_{g,i}}{\partial t} + \rho_{ap}\frac{\partial \overline{q_i}}{\partial t}\right) = 0$$

$$\frac{\partial \overline{q_i}}{\partial t} = \frac{\Omega_c D_{c,i}}{r_c^2}(q_i^* - \overline{q_i})$$

Momentum balance

$$-\frac{\partial P}{\partial z} = \frac{150\mu(1-\varepsilon)^2}{\varepsilon^3 d_p^2}u_0 + \frac{1.75(1-\varepsilon)\rho}{\varepsilon^3 d_p}|u_0|u_0$$

Energy balances—two equations model

$$\frac{\partial}{\partial z}\left(\lambda\frac{\partial T_g}{\partial z}\right) - u_0 C_{g,T} C_p\frac{\partial T_g}{\partial z} + \varepsilon R_g T_g\frac{\partial C_{g,T}}{\partial t} - \frac{4h_w}{d_{wi}}(T_g - T_w) - \left[\varepsilon C_{g,T} C_V + (1-\varepsilon)\left(\varepsilon_p\sum_{i=1}^{n} C_{g,i}C_{v,i} + \rho_p\sum_{i=1}^{n}\overline{q_i}C_{v,\mathrm{ads},i} + \rho_p\hat{C}_{ps}\right)\right]\frac{\partial T_g}{\partial t}$$

$$+(1-\varepsilon)\varepsilon_p R_g T_g\frac{\partial C_{g,T}}{\partial t} + \rho_b\sum_{i=1}^{n}(-\Delta H_{\mathrm{ads}})_i\frac{\partial \overline{q_i}}{\partial t} = 0$$

$$\rho_w\hat{C}_{p,w}\frac{\partial T_w}{\partial t} = \alpha_w h_w(T_g - T_w) - \alpha_{w\ell}U(T_w - T_\infty)$$

Energybalances—one equation model

$$\frac{\partial}{\partial z}\left(\lambda\frac{\partial T}{\partial z}\right) - u_0 C_{g,T} C_p\frac{\partial T}{\partial z} + \varepsilon R_g T\frac{\partial C_{g,T}}{\partial t} - \left[\varepsilon C_{g,T} C_V + (1-\varepsilon)\left(\varepsilon_p\sum_{i=1}^{n} C_{g,i}C_{v,i} + \rho_p\sum_{i=1}^{n}\overline{q_i}C_{v,\mathrm{ads},i} + \rho_p\hat{C}_{ps}\right) + \frac{4\rho_w\hat{C}_{p,w}}{\alpha_w d_{wi}}\right]\frac{\partial T}{\partial t}$$

$$+(1-\varepsilon)\varepsilon_p R_g T\frac{\partial C_{g,T}}{\partial t} + \rho_b\sum_{i=1}^{n}(-\Delta H_{\mathrm{ads}})_i\frac{\partial \overline{q_i}}{\partial t} - \frac{4\alpha_{w\ell}U}{\alpha_w d_{wi}}(T - T_\infty) = 0$$

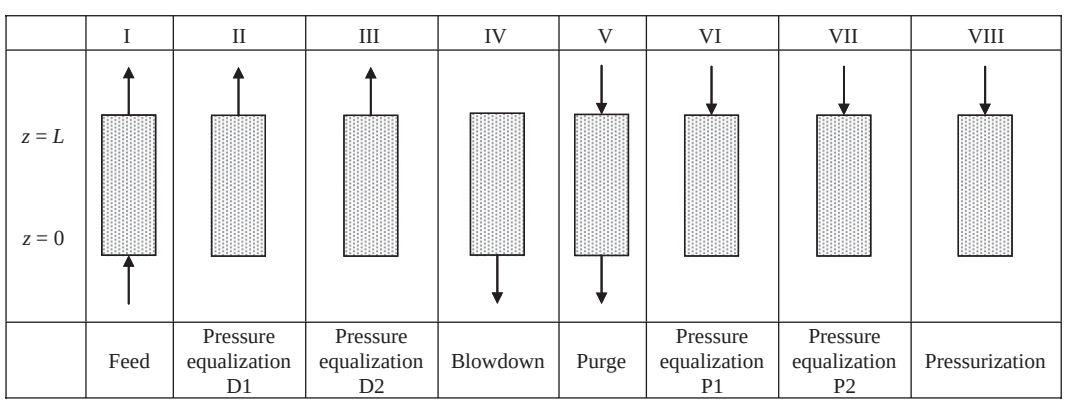


**Fig. 2.** Cycle sequence used in the PSA simulations.

derivatives and integrals. The single column simulations started with the feed step considering that the column was filled with hydrogen at the feed temperature and pressure. In the case of the four column simulation, the first step of each column followed the scheme of Fig. 3 and as initial conditions the columns were also presumed filled with hydrogen at the feed temperature, but with each column at the zero time pressure of the corresponding first step. The implemented multicolumn program solves the model simultaneously for all columns.

### 3.1. Single column results

#### 3.1.1. Complete model

In this section, the simulation results obtained for the single column PSA process using the complete mathematical model are presented. First, the general behaviour of the system will be discussed considering the results of one simulation. Then, we will evaluate the influence of several operating conditions on the PSA performance.

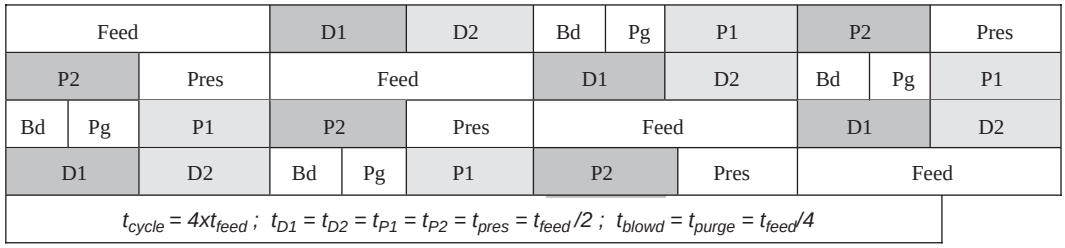


**Fig. 3.** Four columns arrangement with continuous feed using the cycle sequence proposed in Fig. 2.

**Table 4**
Fitting parameters of the multisite Langmuir model for the activated carbon and zeolite

| Gas | $q_{max}$ (mol/kg) | $a_i$ (–) | $K_\infty \times 10^{11}$ ($Pa^{-1}$) | (–ΔH) (kJ/mol) |
|---|---|---|---|---|
| *Activated carbon* | | | | |
| $CO_2$ | 7.8550 | 3.0 | 2.125 | 29.084 |
| $H_2$ | 23.565 | 1.0 | 7.233 | 12.843 |
| $CH_4$ | 6.7329 | 3.5 | 7.904 | 22.701 |
| CO | 9.0634 | 2.6 | 2.680 | 22.577 |
| $N_2$ | 5.8913 | 4.0 | 23.46 | 16.263 |
| *Zeolite* | | | | |
| $CO_2$ | 4.525 | 2.2 | 11.11 | 35.965 |
| $H_2$ | 9.954 | 1.0 | 50.76 | 9.2309 |
| $CH_4$ | 4.976 | 2.0 | 35.65 | 20.643 |
| CO | 3.828 | 2.6 | 3.937 | 29.773 |
| $N_2$ | 4.148 | 2.4 | 30.83 | 20.413 |

As mentioned before, the currently employed technology for the production of hydrogen through cyclic adsorption processes uses two or more adsorbents placed in successive layers in the bed. This kind of arrangements is normally employed to enhance the process performance and reduce unit size and energetic costs.

In this work a layered bed of activated carbon/zeolite was employed. The adsorbents considered are two commercial samples for which the physical properties, as well as, the kinetic and equilibrium data have been determined previously (Lopes et al., 2007). The adsorption equilibrium was represented by the multisite Langmuir model (Eq. (1)) and the corresponding parameters are given in Table 4:

$$\left(\frac{q_i}{q_{i,\max}}\right) = a_i K_i P\left[1 - \sum_i \left(\frac{q_i}{q_{i,\max}}\right)\right]^{a_i} \quad (1)$$

The column dimensions, the physical properties of the adsorbents and the general operating conditions considered in the first simulation are shown in Table 5. An adsorption bed with a length of 1 m was considered. Constant and equal bed porosity on both adsorbent layers were assumed in the simulations. The higher and lower pressures of the process were fixed at 7 and 1 bar, respectively, and the feed temperature was 303 K. As a first approach, the length of the activated carbon layer was considered to be 0.5 m (zeolite layer also with 0.5 m). The feed flow rate was 12.2 N $m^3$/h. This value was established based on the space time of approximately 20 s used in previous studies of layered bed PSA for $H_2$ recovery (Baksh and Ackey, 2002; Baksh and Terbot, 2003; Yang and Lee, 1998). The feed stream was a five component mixture with $CO_2$, $H_2$, $CH_4$, CO and $N_2$. Pure hydrogen was used in the purge and pressurization steps. The purge flow rate used was 3.5 N $m^3$/h, that is, the inlet gas velocity used in the purge step is equal to two times the inlet gas velocity of the feed step. This corresponds to a purge to feed ratio of 0.097, defined as the ratio between the amounts of hydrogen in the purge to that in the feed (Yang, 1987). The feed step duration was 40 s which automatically sets, through the relations given in Fig. 3, the pressure equalization and pressurization steps time to 20 s and the blowdown and purge duration to 10 s.

**Table 5**
Column characteristics, physical properties of the adsorbents and operating conditions of the PSA simulations

| Adsorbent | Layered bed of AC/Zeolite |
|---|---|
| Column length (m) | 1 |
| Activated carbon length (m) | 0.5 |
| Column diameter (m) | 0.2 |
| Wall density (kg/$m^3$) | 8238 |
| Wall specific heat (J/kg/K) | 500 |
| $\alpha_w$ (1/m) | 195 |
| $\alpha_{w\ell}$ (1/m) | 197 |
| Bed porosity | 0.38 |
| Particle porosity | AC: 0.566; Zeo: 0.503 |
| Particle density (kg/$m^3$) | AC: 842; Zeo: 1126 |
| Particle specific heat (J/kg/K) | AC: 709; Zeo: 920 |
| Particle radius (m) | AC: $1.17 \times 10^{-3}$; Zeo: $0.85 \times 10^{-3}$ |
| Feed temperature (K) | 303 |
| Feed pressure (bar) | 7 |
| Feed molar fraction (%) | $CO_2$:16.6; $H_2$:73.3; $CH_4$:3.5; CO:2.9; $N_2$:3.7 |
| Feed flow rate (N $m^3$/h) | 12.2 |
| Purge pressure (bar) | 1 |
| Purge flow rate (N $m^3$/h) | 3.5 |
| Feed duration (s) | 40 |

**Table 6**
Transport parameters values used in the simulations

| | Activated carbon | Zeolite |
|---|---|---|
| $D_p$ ($m^2$/s)[a] | $CO_2$: $3.43 \times 10^{-6}$ | |
| | $H_2$: $4.99 \times 10^{-6}$ | |
| | $CH_4$: $3.06 \times 10^{-6}$ | |
| | CO: $3.01 \times 10^{-6}$ | |
| | $N_2$: $3.02 \times 10^{-6}$ | |
| $D_c/r_c^2$ ($s^{-1}$)[a] | $CO_2$: $1.24 \times 10^{-2}$ | $CO_2$: $1.87 \times 10^{-4}$ |
| | $H_2$: $8.89 \times 10^{-2}$ | $H_2$: $9.23 \times 10^{-2}$ |
| | $CH_4$: $3.96 \times 10^{-3}$ | $CH_4$: $1.04 \times 10^{-2}$ |
| | CO: $2.11 \times 10^{-2}$ | CO: $4.22 \times 10^{-3}$ |
| | $N_2$: $2.29 \times 10^{-2}$ | $N_2$: $2.13 \times 10^{-2}$ |
| $k_f$ (m/s) | $9.28 \times 10^{-2}$ | |
| $h_f$ (W/$m^2$K) | 219 | |
| $h_w$ (W/$m^2$K) | 94 | |
| $U$ (W/$m^2$K) | 0 | |

[a]Values at feed inlet conditions (7 bar and 303 K).

The transport parameter values required in the mathematical model are summarized in Table 6. The macropore diffusivity took into account only the molecular diffusivities which were calculated with the Chapman–Enskog equation (Bird et al., 2002). A tortuosity factor of 2 was considered for both adsorbents. The axial mass and heat dispersion coefficients, as well as, the mass transfer and heat convective coefficient were estimated using the Wakao and Funazkri (1978) correlations (Da Silva, 1999; Yang, 1987). The internal convective heat transfer coefficient between gas and the wall column was calculated with the Wasch and Froment (1972) correlation. The

process was considered adiabatic, i.e. $U=0$. General properties of the gases, like density, viscosity, and molar specific heat were obtained according to Bird et al. (2002). The specific heat and the viscosity of the gas were estimated at the inlet conditions and taken as constants throughout the bed. The molar specific heat of the adsorbed gas was assumed to be equal to the molar specific heat in the gas phase (Sircar, 1985).

The performance of a PSA process is commonly evaluated according to basic parameters of product purity, product recovery and productivity (Rota and Wankat, 1990) defined, respectively, by

$$\text{PURITY} = \frac{\int_0^{t_{\text{feed}}} C_{H_2} u_0|_{z=L}\,\mathrm{d}t}{\sum_{i=1}^{n} \int_0^{t_{\text{feed}}} C_i u_0|_{z=L}\,\mathrm{d}t} \tag{2}$$

$$\text{RECOVERY} = \frac{\int_0^{t_{\text{feed}}} C_{H_2} u_0|_{z=L}\,\mathrm{d}t - \int_0^{t_{\text{press}}} C_{H_2} u_0|_{z=L}\,\mathrm{d}t - \int_0^{t_{\text{purge}}} C_{H_2} u_0|_{z=L}\,\mathrm{d}t}{\int_0^{t_{\text{feed}}} C_{H_2} u_0|_{z=0}\,\mathrm{d}t} \tag{3}$$

$$\text{PRODUCTIVITY} = \frac{\left(\int_0^{t_{\text{feed}}} C_{H_2} u_0|_{z=L}\,\mathrm{d}t - \int_0^{t_{\text{press}}} C_{H_2} u_0|_{z=L}\,\mathrm{d}t - \int_0^{t_{\text{purge}}} C_{H_2} u_0|_{z=L}\,\mathrm{d}t\right) A_{\text{bed}}}{t_{\text{total}} w_{\text{ads}}} \tag{4}$$

The values obtained for this simulation were:

Hydrogen recovery: 51.84%.
Hydrogen purity: 99.9994%.
Productivity: 59.6 $\text{mol}_{H_2}/\text{kg}_{\text{ads}}/\text{day}$.

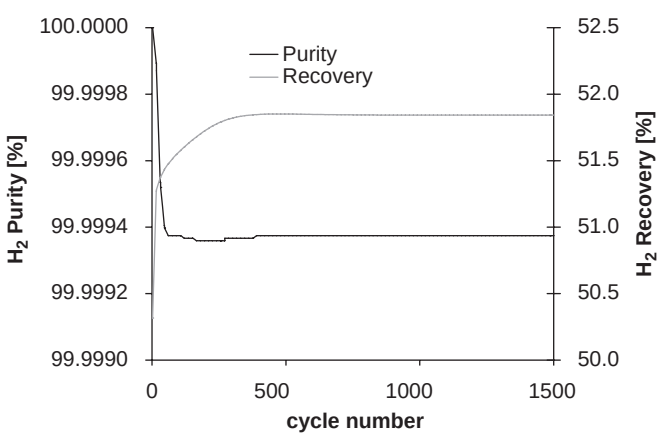


**Fig. 4.** Evolution of the hydrogen purity and recovery values with the simulation cycle number.

The evolution of hydrogen purity and recovery as a function of the simulation cycle number is shown in Fig. 4. It can be seen that CSS is achieved only after 500 cycles. The main reason of this delay in achieving CSS is the stabilization of the temperature profiles within the whole bed. This can be seen in Fig. 5(a), where the gas temperature profiles at the end of the feed step are shown for different cycle numbers. On the other hand, the gas phase concentration profiles of all impurities reach an apparent CSS much faster. As an example, the methane gas phase profiles at the end of the feed step for the same cycle numbers are shown in Fig. 5(b) where no change is observed after the 105th cycle.

The pressure and molar flow rates obtained at the column outlet during one cycle in CSS are shown in Fig. 6. It can be seen that during the pressure equalization steps a considerable amount of hydrogen is "recycled" which improves the recovery of the process. The amount recycled represents approximately 79% of the amount produced. Nevertheless a significant amount of hydrogen is still lost in the blowdown step (over 37% of the amount produced) and also part of the product must be used to repressurize the column.

The impurities observed at the feed outlet stream were $CH_4$ and $N_2$, respectively, in the relative amounts of 0.0005% and 0.0001%. The high diffusion resistance of methane on the activated carbon (low $D_c/r_c^2$) is the reason why methane elutes to the bed end. The activated carbon, having higher adsorption capacity for methane than the zeolite, should retain it. But the low diffusivity value makes both the uptake and regeneration more difficult which broadens the $CH_4$ profiles along the bed. This can be seen in the next figures where the gas (Fig. 7) and solid (Fig. 8) phase concentration profiles obtained along the bed at the end of each step at CSS are presented. Comparing the diffusivity time constant of methane in this activated carbon with values reported in the literature (Jee et al., 2001; Park et al., 1998, 2000) it is observed that the value used in this study is almost one order of magnitude lower. The only exception is a value reported by Warmuzinski and Tanczyk (1997) where a value of the same order of magnitude as the one determined in our experiments

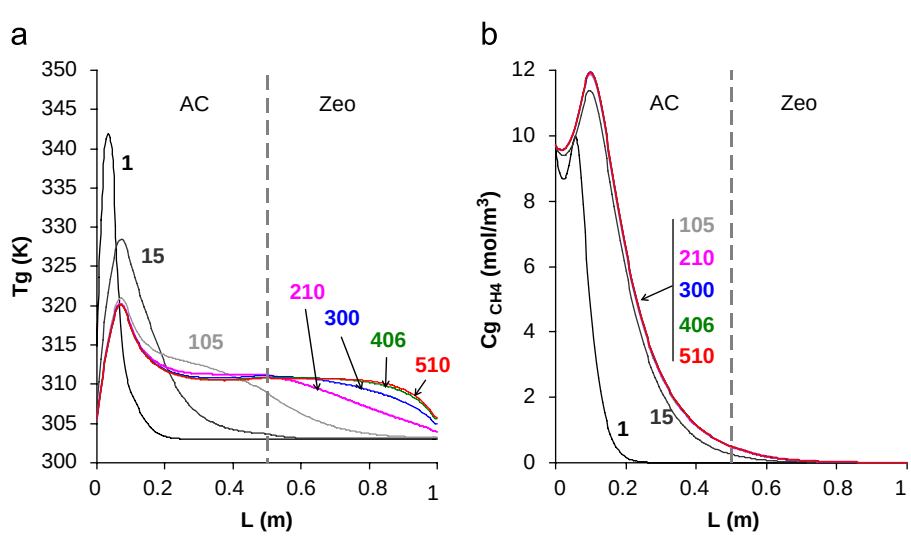


**Fig. 5.** Evolution of the (a) gas temperature and (b) methane gas phase concentration profiles at the end of the feed step with the cycle number.

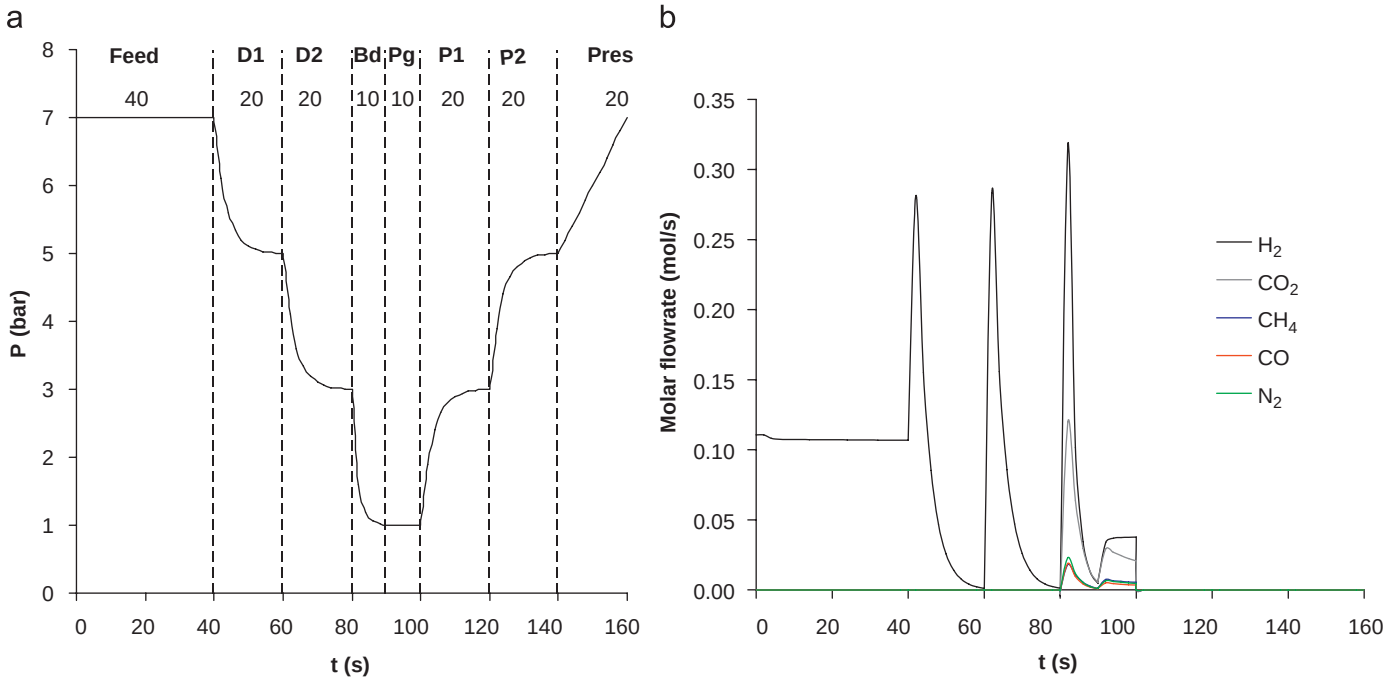


**Fig. 6.** (a) Pressure and (b) molar flowrates obtained at the column outlet at cyclic steady state.

was reported. With this value of methane diffusion within the activated carbon sample, the adsorption of this gas is not limited by the equilibrium capacity, but by kinetic effects.

Figs. 7 and 8 also show that carbon dioxide does not reach the zeolite layer, which, as mentioned above, is one of the design requisites for this process. In addition, the carbon monoxide stays well within the bed as it is strongly adsorbed in the zeolite, but also poorly regenerated. On the other hand, the difference in capacities between the two adsorbents for $N_2$ is very small and this is also illustrated in the bed profiles. As one of the duties of the second layer in the bed is to preferentially retain nitrogen, an adsorbent with higher capacity for $N_2$ should be used.

Fig. 9 shows the temperature profiles obtained along the bed at the end of each step at CSS. Maximum temperature oscillations of around 15 K are observed in some axial positions near the entrance of the activated carbon layer. The temperature increases during adsorption and decreases when the adsorbent is being regenerated. In the zeolite layer, no significant temperature variations are seen in the entire cycle. This is a consequence of the small variations in the adsorbed phase concentration observed during the cycle as discussed above and shows that the benefits of using the zeolite layer are small.

The effect of the following operating parameters on the PSA performance were studied: feed flow rate, purge to feed ratio, activated carbon length at constant total length, activated carbon length at constant zeolite length and zeolite length at constant activated carbon length. The results obtained are shown in Fig. 10. The hydrogen purity decreases when the feed flow rate is increased (graphic (a)), when the purge to feed ratio is decreased (graphic (b)) and also if shorter beds are used either by decreasing the activated carbon layer (graphic (d)) or the zeolite layer (graphic (e)). In all cases, as the purity decreases the process recovery increases. On the other hand, a different trend is observed when the activated carbon length is changed at a constant total length (graphic (c)). In this case, starting with a bed completely filled with activated carbon, both purity and recovery improve as the zeolite layer increases until a limiting value after which the typical tendency is again observed. This inversion occurs when the activated carbon length is no longer sufficient to retain the carbon dioxide. The carbon dioxide enters the zeolite layer and, as it has a very favourable isotherm in this adsorbent, the regeneration is extremely difficult just by lowering the pressure: it starts to accumulate in the zeolite layer and "pushes" the other impurities towards the bed end and eventually also elutes through the bed. Note that all the points in this figure were obtained from simulations stopped at the 1500th cycle. That is, for the simulation corresponding to $L_{AC}/L = 0.1$ the complete CSS was not achieved as the $CO_2$ is still slightly propagating in the zeolite layer.

In graphic (f) of the same figure, the effects of the different parameters studied on the process performance are compared. As mentioned above, both the increase in flow rate and the decrease in purge to feed ratio induce a decrease in purity and also an increase in recovery. However, from this graphic, it can be seen that, when comparing the relative influence of these two parameters, higher recoveries are obtained for the same purity with the change in the feed flow rate. It can also be seen that the influence of the activated carbon length at constant total length is minor when compared to the influence of the other parameters. On the other hand, the figure also shows that higher recovery for the same purity is observed when the activated carbon length is reduced at constant zeolite length.

As mentioned above, the value of the methane micropore diffusivity on the activated carbon obtained experimentally by Lopes et al. (2007) and used in this study (Table 6) is one order of magnitude lower than the value most frequently used in other published hydrogen purification studies. Thus, simulations using a value of methane micropore diffusivity on the activated carbon 10 times greater than the value of Table 6 were also performed for comparison purposes. The results obtained are shown in graphics (a)–(e) of Fig. 10 as dashed lines. It can be seen that, although the process recovery values do not change significantly, the hydrogen purity is strongly affected. That is, if the mass transfer resistance of methane on the activated carbon is lower, a purer hydrogen stream can be produced for the same recovery.

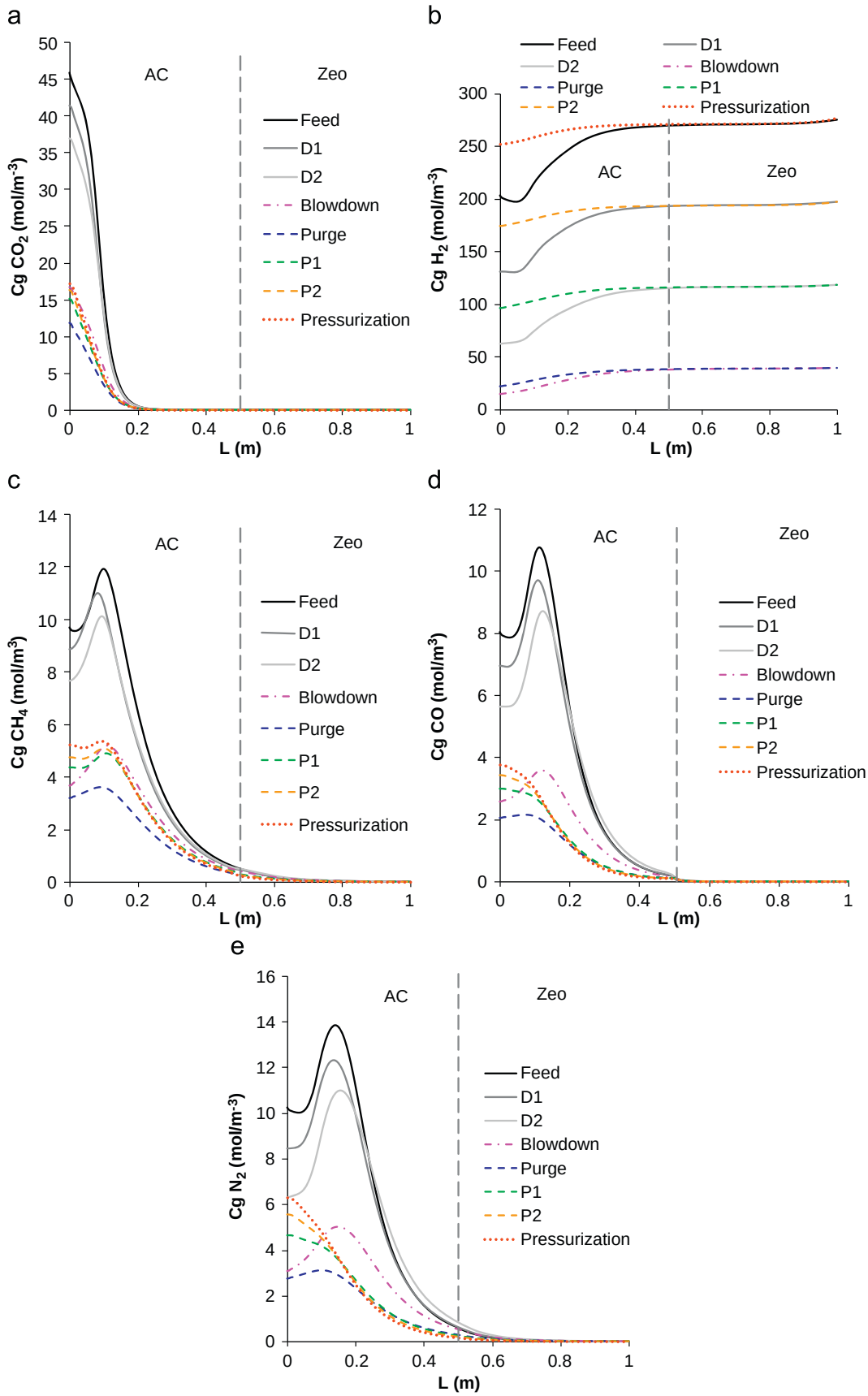


**Fig. 7.** Gas phase concentration profiles obtained along the bed at the end of each step at cyclic steady state: (a) $CO_2$, (b) $H_2$, (c) $CH_4$, (d) CO and (e) $N_2$.

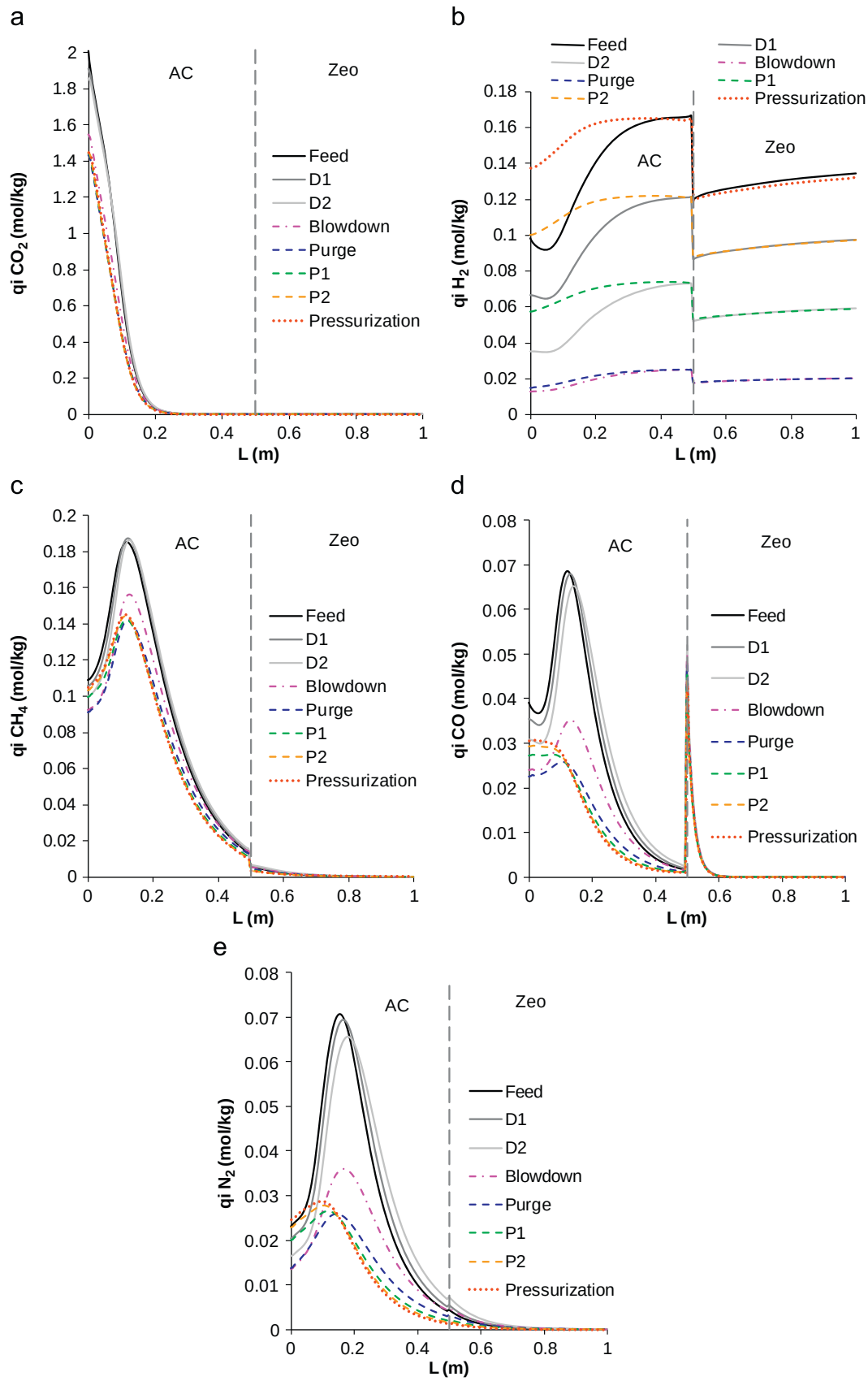


**Fig. 8.** Solid phase concentration profiles obtained along the bed at the end of each step at cyclic steady state: (a) $CO_2$, (b) $H_2$, (c) $CH_4$, (d) CO and (e) $N_2$.

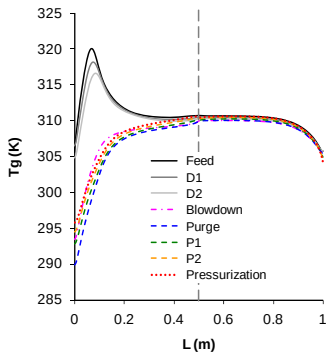


**Fig. 9.** Temperature profiles obtained along the bed at the end of each step at cyclic steady state.

### 3.1.2. Reduced model

In order to assess the validity of the model simplifications proposed in Section 2.2, numerical predictions obtained with the reduced models (Table 3) were compared with those obtained with the complete model (Table 1). The operation and adsorbent parameters given in Tables 4–6 were employed for the three simulations and the obtained results are presented in the following figures. Figs. 11–13(a) show, respectively, the gas phase, solid phase and temperature profiles along the bed at the end of the feed and purge steps at CSS for the complete, the two energy balances reduced model and the one energy balance reduced model.

Looking at these profiles, it can be seen that only small differences are encountered between the results obtained with the complete model and the two energy equations model. The positions of the concentrations and temperature fronts are, for these two models,

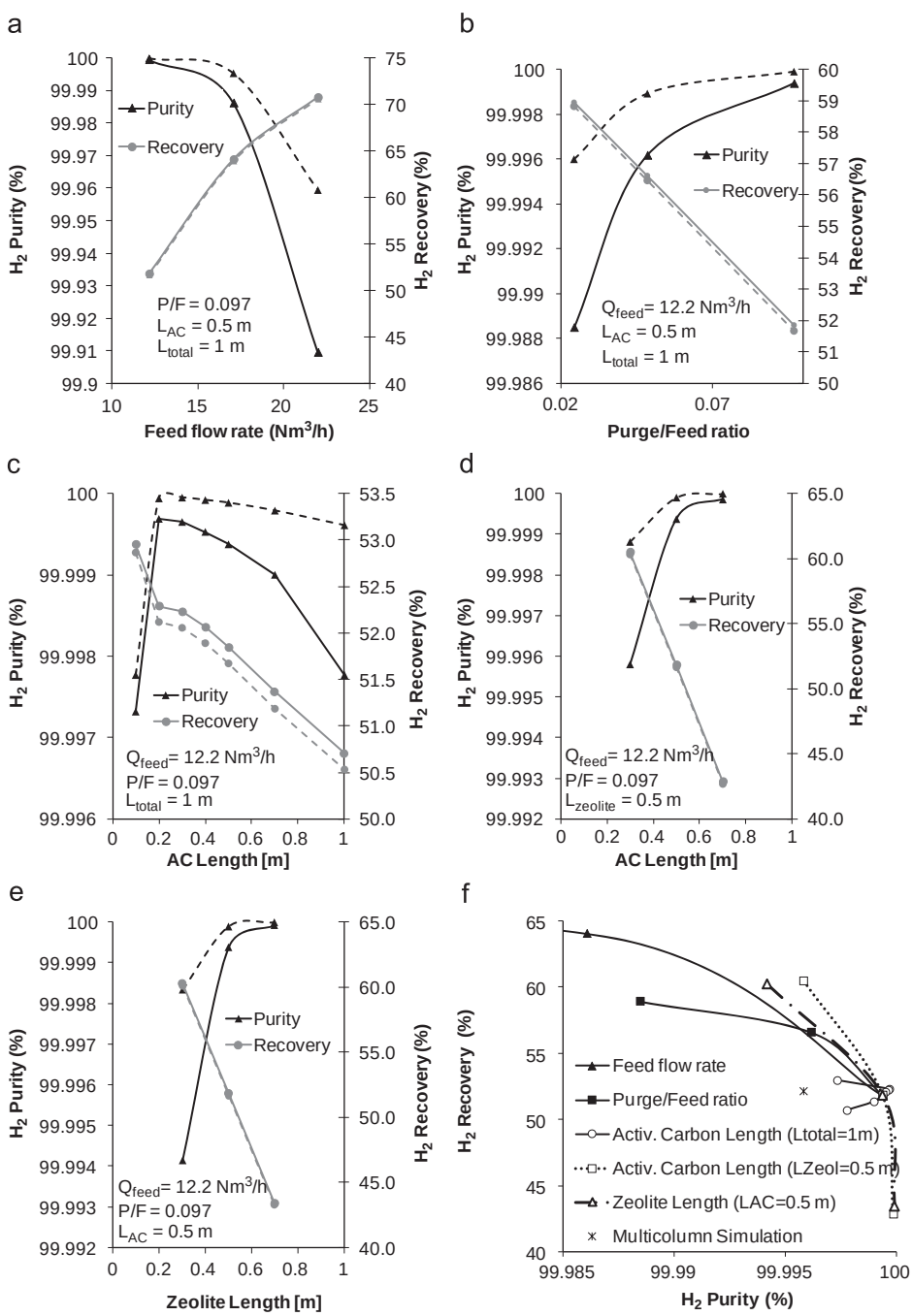


**Fig. 10.**

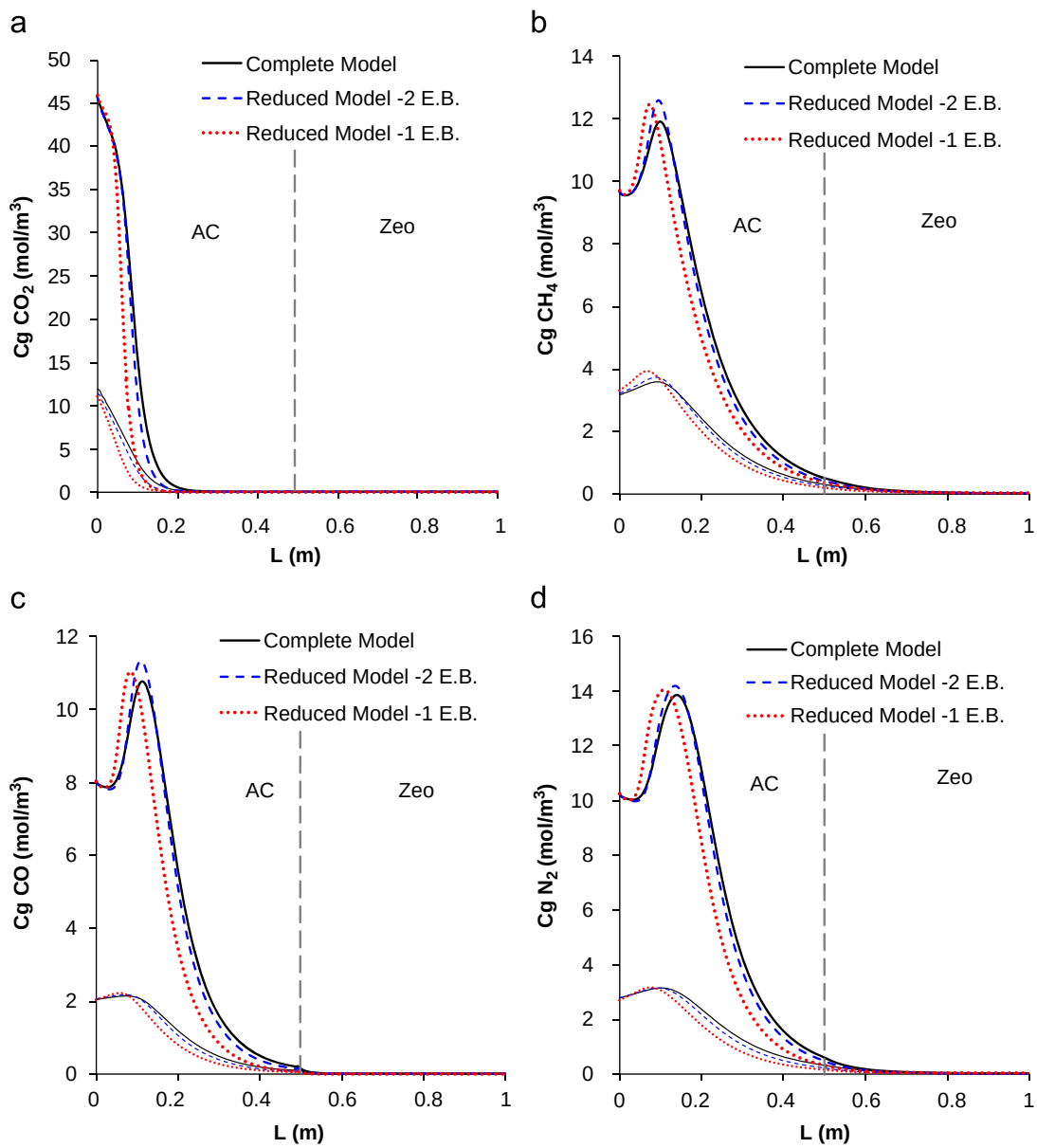


**Fig. 11.** Gas phase concentration profiles obtained along the bed at the end of the feed (——) and purge (–) steps at cyclic steady state for the complete and reduced models: (a) $CO_2$, (b) $CH_4$, (c) CO and (d) $N_2$.

almost overlapped and only the peaks' height differs slightly. On the other hand, if the energy balance is further simplified, assuming complete thermal equilibrium within the packed bed system, the profiles become noticeably different. A temperature difference of over 5 K is observed throughout the entire column and also the fronts' positions are closer to the bed inlet.

In Fig. 13(b) a comparison between the gas, solid and wall temperature profiles at the end of the feed and purge steps at CSS obtained with the complete model is made. It is shown that the gas and solid temperature profiles coincide, while a significant difference to the column wall temperature exists.

The performance parameters predicted with these simulations were, respectively, 51.90% hydrogen recovery and 99.9996% hydrogen purity for the two energy balances reduced model; and 50.53% hydrogen recovery and 99.9997% hydrogen purity for the one energy balance reduced model. These values compare with 51.84% and 99.9994% reported in the previous section for the complete model.

With these results it can be concluded that the assumptions made for the formulation of the two energy equations model are acceptable. With this reduced model the unknown variables in the simulations are reduced by around 20% when compared to the complete model.

### 3.2. Multicolumn system results

The single column simulations use a pure hydrogen stream for all the repressurization steps (in the pressure equalizations and in the pressurization itself) and also in the purge step. However, in multicolumn arrangements, the stream entering the bed during these steps will not be a "clean" one, but will instead contain impurities

**Fig. 10.** PSA performance: effect of feed flow rate (a), purge to feed ratio (b) and activated carbon length at constant total length (c), activated carbon length at constant zeolite length (d) and zeolite length at constant activated carbon length (e) on the obtained hydrogen purity and recovery ((—) micropore diffusivity values of Table 6, (- - -) micropore diffusivity of methane one order of magnitude higher than that of Table 6); and comparison between the effect of the different operating parameters studied and the multicolumn simulation (f).

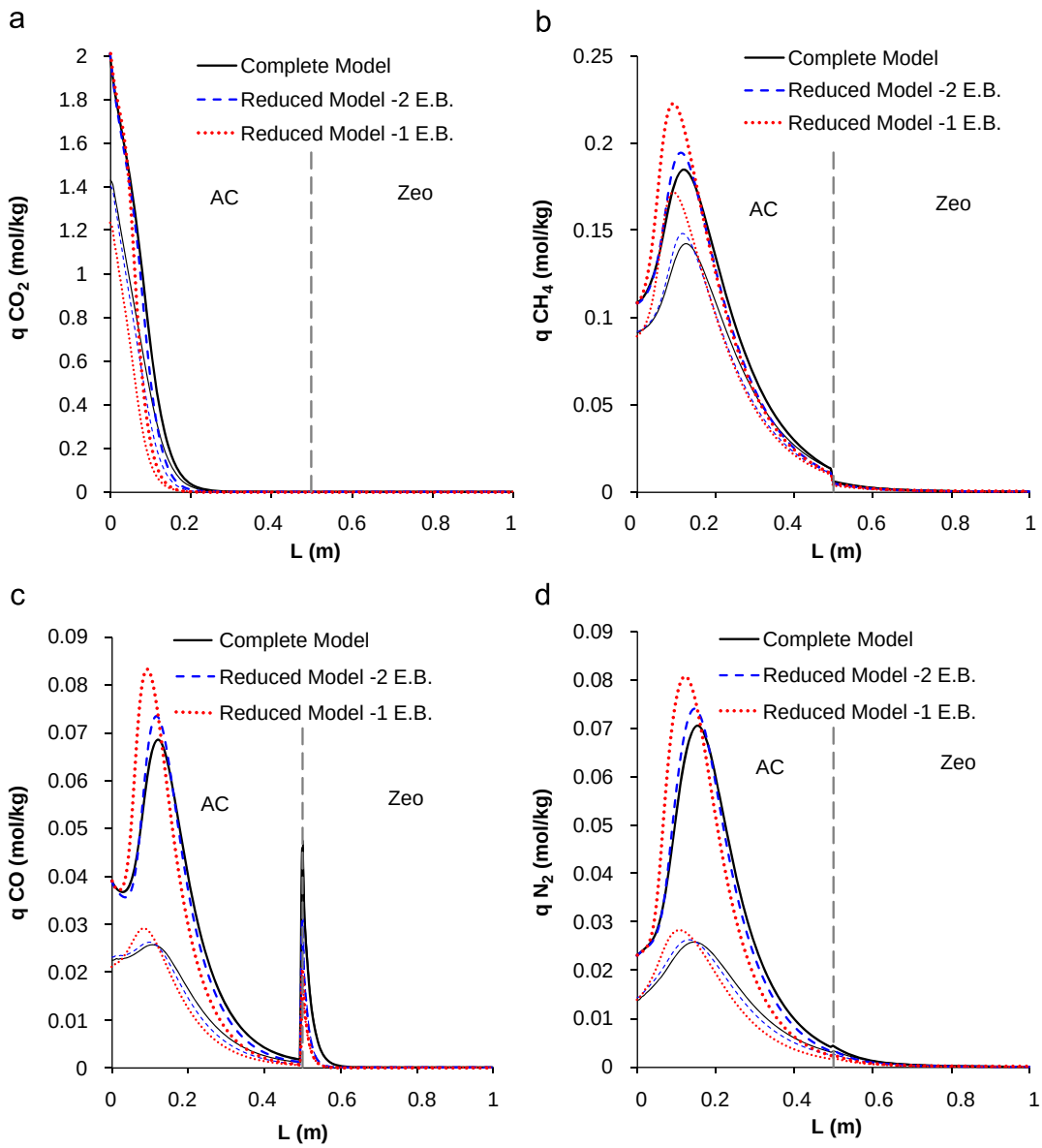


**Fig. 12.** Solid phase concentration profiles obtained along the bed at the end of the feed (——) and purge (–) steps at cyclic steady state for the complete and reduced models: (a) $CO_2$, (b) $CH_4$, (c) CO and (d) $N_2$.

that are leaving another column in the process. This is a more realistic representation of actual PSA operation. Therefore, the complete mathematical model presented above was applied to the simulation of a four column PSA process in accordance with the column schedule given in Fig. 3.

In the particular case of hydrogen, where product purities above 99.99% are required, the purge and pressurization streams are almost pure and thus the single column simulations are not far from reality. They allow a fastest route to search for the appropriate operating conditions that meet the process specifications. Of course even a small amount of impurities in the recycled streams has some effect on the process performance which must be evaluated. Thus a multicolumn simulation using the same operating and adsorbent parameters (Tables 4–6) that resulted in a high purity product was performed. A hydrogen recovery, purity and productivity, respectively, of 52.11%, 99.9958% and 59.9 $mol_{H_2}/kg_{ads}$/day were obtained, with $CH_4$ and $N_2$ present as impurities at the end of the feed step in the relative amounts of 0.0029% and 0.0013%. The performance of the multicolumn simulation is compared in Fig. 10(f) with the results previously determined for the single column process, showing the inferior performance of the multicolumn process, that is, for the same recovery, the purity of the obtained product is poorer. As mentioned above the worsening of the process performance is due to the "dirty" streams used in the pressurization and purge steps which lead to the accumulation of impurities near the bed end. This is illustrated in Fig. 14 where the evolution of the gas and solid phase concentrations of $N_2$ during the course of one cycle at the end of the column is shown for both the single and multicolumn simulations as an example.

Nevertheless and as pointed out before, with such high purity specification, the influence of the recycling stream is minor. In the case illustrated above, the product purity decreases only from 99.9994% in the single column simulation to 99.9958% in the multicolumn simulation. However, if operating conditions that originate less pure recycling streams are considered, the effect is much more important. This can be seen from the results of a multicolumn

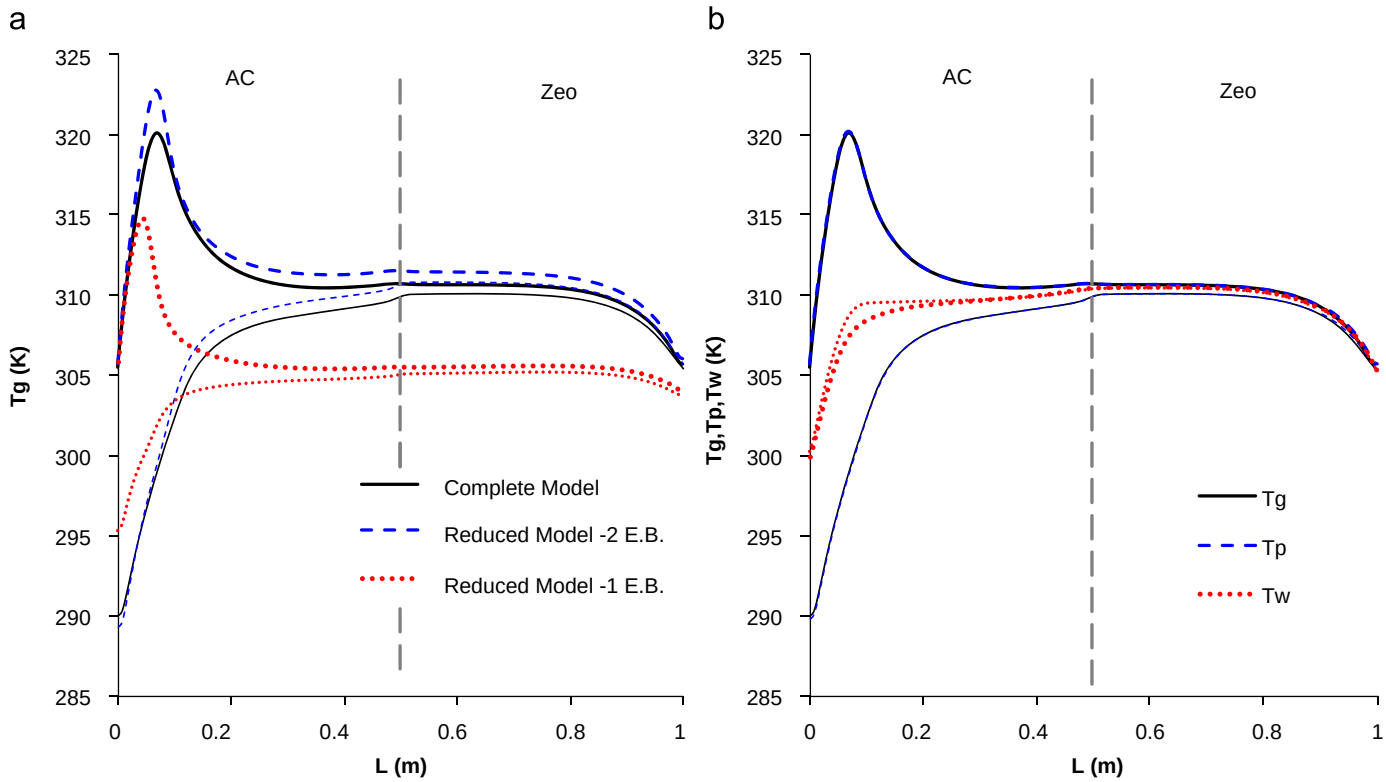


**Fig. 13.** Temperature profiles obtained along the bed at the end of the feed (——) and purge (–) steps at cyclic steady state: (a) comparison between the results of the complete and reduced models; (b) comparison between the three temperatures in the complete model.

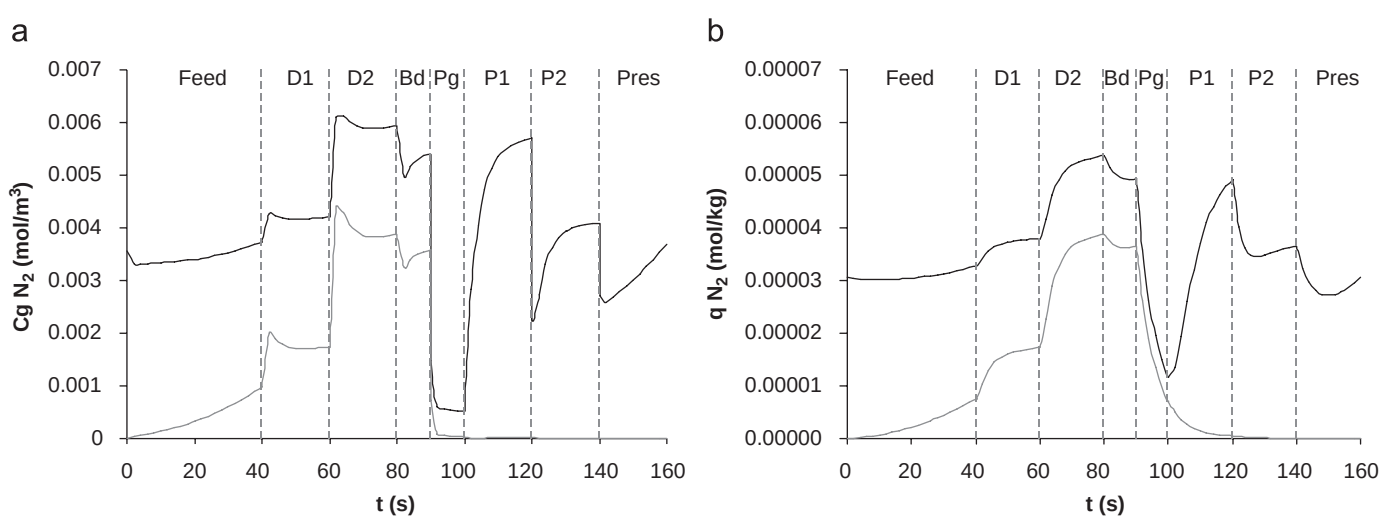


**Fig. 14.** Nitrogen gas (a) and solid (b) phase concentrations at $L = 1$ m during one cycle obtained for the four columns (–) and single column (——) simulations.

simulation using a feed flow rate of 17.1 N $m^3$/h. For this simulation a hydrogen purity of 99.9256% is obtained which compares to a value of 99.9860% obtained in the single column simulation. In this case a larger difference is observed between the two simulations.

## 4. Conclusions

The purification of hydrogen by a layered activated carbon/zeolite PSA process from a feed stream mixture containing as impurities $CO_2$, $CH_4$, CO and $N_2$ was studied by simulation. The complete model able to describe the dynamic behaviour of the pressure swing adsorption system was presented. The model was applied in the simulation of a PSA process with an eight steps cycle scheme.

In a single column simulation, a 99.9994% purity hydrogen stream was attained at the end of the feed step for a process hydrogen recovery of 51.84% and a productivity of 59.6 $mol_{H_2}/kg_{ads}$/day. Both methane and nitrogen were encountered as impurities at the end of the feed step. The study of the influence of the feed flow rate, the purge to feed ratio and the length of the activated carbon layer on the process performance revealed that, within the ranges of parameters

values studied, the best performance is obtained with the change of the feed flow rate. It also showed that the introduction of the zeolite layer improves, although slightly, both the purity and recovery of the process until a limiting value after which the activated carbon layer becomes too small to retain the carbon dioxide.

An extension of the cycle under consideration was proposed for a four column process. The multicolumn simulation showed that the columns interconnection, which takes into account the impurities in the streams used for repressurization and purge, results in a poorer process performance. With this simulation a hydrogen recovery, purity and productivity of, respectively, 52.11%, 99.9958% and 59.9 $mol_{H_2}/kg_{ads}$/day were obtained.

The complete model was simplified based on the assumption that one of the mass transfer mechanisms controls and also assuming thermal equilibrium between system phases and reduced models were formulated. The reduced model predictions were evaluated by comparing its results with the ones obtained from the complete model. It is shown that the model that merely takes into account the micropore resistance (described by the LDF model) and assumes thermal equilibrium between the gas and the adsorbent satisfactorily predicts the behaviour of the pressure swing adsorption unit.

## Notation

| | |
|---|---|
| $a_p$ | particle specific area, $m^{-1}$ |
| $a_i$ | number of neighbouring sites occupied by the adsorbate molecule in the multisite Langmuir isotherm, dimensionless |
| $A_{bed}$ | bed area, $m^2$ |
| $Bi_i$ | mass Biot number of component $i$ ($Bi_i = a_p k_f R_p^2/\varepsilon_p \Omega_M D_{p,i}$), dimensionless |
| $C_{g,i}$ | gas phase concentration of component $i$, $mol/m^3$ |
| $C_{g,T}$ | total gas phase concentration, $mol/m^3$ |
| $C_p$ | gas mixture molar specific heat at constant pressure, J/mol K |
| $\overline{C_{p,i}}$ | average concentration of component $i$ in the macropores, $mol/m^3$ |
| $\hat{C}_{ps}$ | particle specific heat at constant pressure (per mass unit), J/kg K |
| $\overline{C_{p,T}}$ | average total concentration in the macropores, $mol/m^3$ |
| $\hat{C}_{pw}$ | wall specific heat at constant pressure (per mass unit), J/kg K |
| $C_v$ | gas mixture molar specific heat at constant volume, J/mol K |
| $C_{v,ads,i}$ | molar specific heat of component $i$ in the adsorbed phase at constant volume, J/mol K |
| $C_{v,i}$ | molar specific heat of component $i$ at constant volume, J/mol K |
| $d_p$ | particle diameter, m |
| $d_{wi}$ | internal bed diameter, m |
| $D_{ax}$ | axial dispersion coefficient, $m^2/s$ |
| $D_{c,i}$ | micropore diffusivity of component $i$, $m^2/s$ |
| $D_{p,i}$ | macropore diffusivity of component $i$, $m^2/s$ |
| $h_f$ | film heat transfer coefficient between the gas and particle, $J/s\,m^2\,K$ |
| $h_w$ | film heat transfer coefficient between the gas and wall, $J/s\,m^2\,K^1$ |
| $(\Delta H_{ads})_i$ | heat of adsorption of component $i$, J/mol |
| $k_f$ | film mass transfer coefficient, m/s |
| $k_s$ | geometrical factor (0—slab; 1—cylinder; 2—sphere), dimensionless |
| $K_i$ | equilibrium constant in the multisite Langmuir isotherm, Pa |
| $L$ | bed length, m |
| $P$ | pressure, Pa |
| $\overline{q_i}$ | particle averaged adsorbed concentration, mol/kg |
| $q_{i,max}$ | specific saturation adsorption capacity in the multisite Langmuir isotherm, mol/kg |
| $q_i^*$ | adsorbed concentration in equilibrium with $\overline{C_{p,i}}$, mol/kg |
| $Q_{feed}$ | feed flow rate, $N\,m^3/h$ |
| $r_c$ | "microparticle" radius, m |
| $R_g$ | ideal gas constant, J/mol K |
| $R_p$ | particle radius, m |
| $t$ | time, s |
| $T_g$ | bulk phase temperature, K |
| $T_p$ | solid temperature, K |
| $T_w$ | wall temperature, K |
| $T_\infty$ | ambient temperature, K |
| $u_0$ | superficial velocity, m/s |
| $U$ | overall heat transfer coefficient, $J/s\,m^2\,K$ |
| $w_{ads}$ | mass of adsorbent, kg |
| $y_i$ | component I molar fraction, dimensionless |
| $z$ | axial position, m |

*Greek letters*

| | |
|---|---|
| $\alpha_w$ | ratio of the internal surface area to the volume of the column wall, $m^{-1}$ |
| $\alpha_{w\ell}$ | ratio of the log mean surface to the volume of column wall, $m^{-1}$ |
| $\varepsilon$ | bed porosity, dimensionless |
| $\varepsilon_p$ | particle porosity, dimensionless |
| $\lambda$ | heat axial dispersion coefficient, J/s m K |
| $\mu$ | bulk gas mixture viscosity, kg/m s |
| $\rho$ | bulk gas mixture density, $kg/m^3$ |
| $\rho_p$ | particle density, $kg/m^3$ |
| $\rho_w$ | wall density, $kg/m^3$ |
| $\Omega$ | LDF factor [$\Omega = (k_s+1)(k_s+3)$], dimensionless |

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